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Marketing Analytics (MGT 7215)

Assignment 2: Product Design

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Introduction

In the contemporary landscape, grasping consumer preferences stands as a fundamental requisite for the development and enhancement of diverse products within specific target markets. Each product encompasses an array of features and functionalities that resonate with individual consumers on various levels. To address this diversity, many companies are expanding their product portfolios to widen their reach among consumers. This endeavor demands meticulous consideration and strategic planning to effectively contend in the market.

Within this framework, conjoint analysis emerges as a potent analytical instrument, enabling analysts to uncover the perceived value of individual product attributes. Initially, analysts determine the buyer's part-worths for attributes, which are subsequently employed to anticipate consumer choices. Furthermore, Principal Component Analysis (PCA) is integrated into this report. PCA primarily serves to reveal underlying patterns influencing consumer behavior (Tsourgiannis & Valsamidis, 2021).

Literature Review

Conjoint analysis is a powerful tool in marketing research, offering valuable insights into consumer behavior (Ding, 2007). It helps researchers understand what features matter most to consumers and how they make choices when presented with different product options (Green & Srinivasan, 1990). Beyond understanding individual preferences, conjoint analysis also sheds light on market dynamics and structures (Rhee & Park, 2011). This makes it a valuable tool for market segmentation and product positioning, allowing businesses to tailor their offerings to specific consumer groups (Green & Krieger, 1991). Additionally, conjoint analysis can be used to predict how consumers might react to new product introductions or improvements to existing products (Pistol & Bucea-Manea-Tonis, 2017). In conjunction with conjoint analysis, perceptual mapping, which involves techniques such as multidimensional scaling (MDS) and principal component analysis (PCA), is used to visualize and understand consumer perceptions of brands in the marketplace (Greenacre, 2009). This method helps in identifying the positioning of products or brands in the minds of consumers, aiding in strategic planning and decision-making (Ganguly et al., 2011). Moreover, perceptual mapping based on web search queries and consumer forum comments has been proposed as a technique to understand market structure and monitor consumers' perceptual changes (Won et al., 2018). In the context of marketing, PCA can be applied to perceptual mapping to simplify the visualization of consumer perceptions and preferences (Greenacre, 2009). Additionally, PCA is utilized to characterize the evolving correlation structures of financial markets and measure associated systemic risk, making it relevant in the analysis of market dynamics (Nobi & Lee, 2016).

Methodology

We've segmented the provided assignment into four distinct stages. First running Conjoint Analysis where the impact of each feature is calculated. Then visualizing see the impact on tableau plot. Afterwards, Principal Component Analysis is calculated in R to calculate the PVE and plot scores for perceptual mapping and finally Perceptual Mapping is done in Tableau.

Conjoint Analysis

A conjoint analysis was conducted to evaluate consumer preferences for a new circular product under development. The analysis used 18 different product profiles, each defined by various levels of six key product attributes.

- Environment Friendliness
- Delivery Time
- Service Level
- Price
- Quality of Material
- Marketing Proficiency

Additionally, another dataset, named "conjoint preferences," has been included, containing ratings provided by 10 customers. Ratings range from 1 to 10, where 10 signifies high satisfaction and 1 represents low satisfaction with the product.

Both the datasets were loaded into R using `read.excel()` and `read.csv()` function.

```
# Read the Excel file and store it in a Laptop_DP
product_profiles <- read.csv(file.choose())
product_profiles <- product_profiles[, -1]

pref_data <- read_excel("C:/Users/Uzair Khan/Downloads/Conjoint Preferences.xlsx")
```

Figure 1: Read Datasets in R

The total number of combinations of product attributes was also determined during the analysis as follows:

$$\begin{aligned} & 3(\text{Environmental Friendliness}) \times 3(\text{Delivery time}) \\ & \times 3(\text{Service Level}) \times 3(\text{Price}) \times 2(\text{Quality of Material}) \\ & \times 2(\text{Marketing Proficiency}) = 324 \end{aligned}$$

However, the "Product Profiles.csv" file contains 18 product profiles selected for the study. These profiles were chosen based on a fractional factorial design, aiming to ensure statistical efficiency. This design approach entails selecting a subset of combinations from the full set of possible combinations generated by the `expand.grid(attrib.level)` function, which would otherwise result in a full factorial design.

The decision to limit the profiles to 18 arises from practical considerations. In conjoint analysis, a full factorial design would entail including all possible combinations of the given attribute levels, resulting in many combinations (324 in this case). Analyzing such a vast number of

profiles could be impractical and time-consuming, potentially diminishing the reliability of the data.

By employing a fractional factorial design, researchers can ensure that the effects of each factor are captured efficiently while reducing the data requirements and survey costs. This approach allows for the capture of essential insights while mitigating respondent fatigue, thereby maintaining the quality and reliability of the data.

The fractional factorial design stands out for its statistical efficiency, as it contains crucial information necessary for estimating the part-worths of each attribute level. It maintains orthogonality, ensuring that each attribute level varies independently without correlation among them. This feature contributes to a clear and unbiased understanding of the attributes' contributions.

Moreover, the 18 selected profiles reflect the options in real-market options, providing a genuine representation of actual consumer choices. This alignment with real-world scenarios enhances the relevance and validity of the study's findings.

Next, to estimate part-worths for each respondent, a loop was applied using the *caPartUtilities()* function on R from the conjoint package. The following table shows the values which were taken as base values for the calculation.

Environmental Friendliness	0% CO2 reduction
Delivery Time	14 days
Service Level	5 Years Warranty
Price	1000 GBP
Quality of Material	Market Average
Marketing Proficiency	Not very proficient and poor communication

Visualizing Scores

To visualize multiple aspects of consumer preferences and market share, several calculated fields were generated in Tableau. Shared below are the calculated fields created for each step.

Average Attributes Part-Worths:

First, a plot was created to view the attributes and how they contribute to the part-worths of the product. A figure is attached below.

Average Attributes Part-Worths

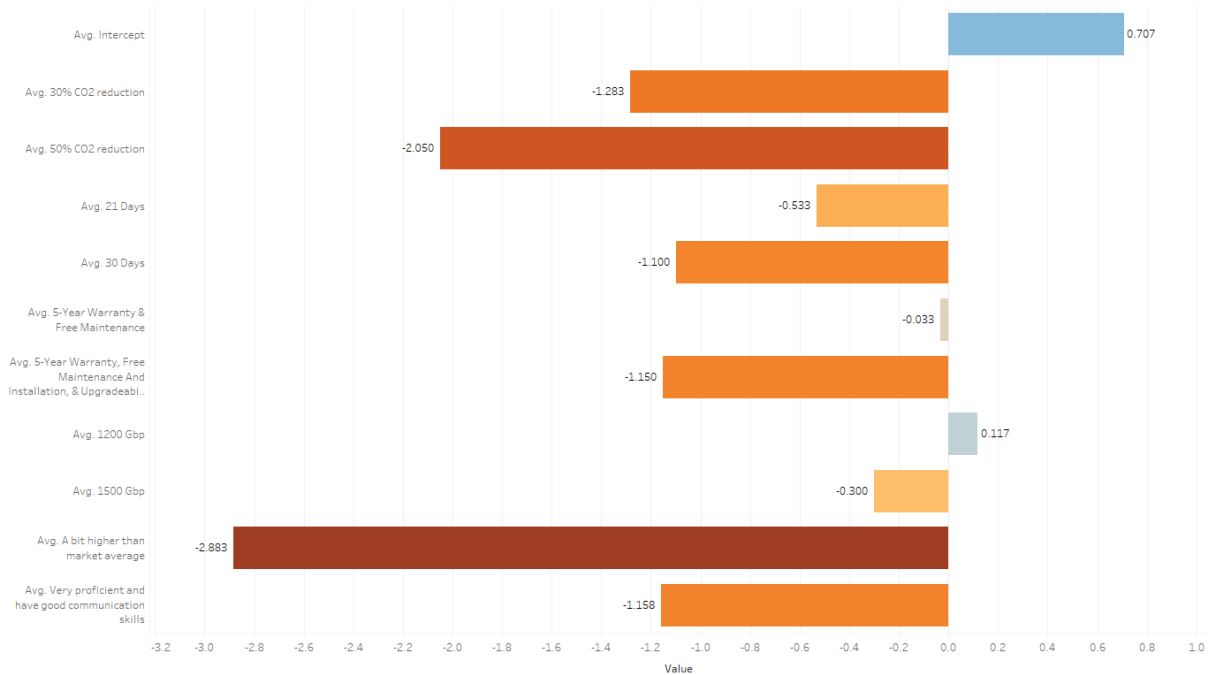


Figure 2: Average Attributes Part-Worths

Attribute Importance Measures:

Next, we went on to calculate the average impact of each feature on the overall product. We calculated the average by using formula:

$$\text{Max Utility of each feature} - \text{Min Utility of each feature}$$

Environmental Friendliness	MAX([0% CO2 reduction], MAX([30% CO2 reduction],[50% CO2 reduction])) - MIN([0% CO2 reduction],MIN([30% CO2 reduction], [50% CO2 reduction]))
Delivery Time	MAX([14 Days], MAX([21 Days],[30 Days])) - MIN([14 Days],MIN([21 Days],[30 Days]))
Service Level	MAX([5-Year Warranty], MAX([5-Year Warranty & Free Maintenance],[5-Year Warranty, Free Maintenance <u>And</u> Installation, & Upgradeability])) - MIN([5-Year Warranty],MIN([5-Year Warranty & Free Maintenance],[5-Year Warranty, Free Maintenance And Installation, & Upgradeability]))
Price	MAX([1000 Gbp], MAX([1200 Gbp],[1500 Gbp])) - MIN([1000 Gbp],MIN([1200 Gbp],[1500 Gbp]))
Quality if Material	MAX([Market average], [A bit higher than market average]) - MIN([Market average], [A bit higher than market average])
Marketing Proficiency	MAX([Not very proficient and poor communication], [Very proficient and have good communication skills]) - MIN([Not very proficient and poor communication], [Very proficient and have good communication skills])

Percentage Attribute Importance Measures:

A similar chart was plotted, however, this time the percentage impact was calculated.

Environmental Friendliness	$\text{AVG}([\text{Environment}]) / (\text{AVG}([\text{Environment}]) + \text{AVG}([\text{Delivery Time}]) + \text{AVG}([\text{Service Level}]) + \text{AVG}([\text{Price}]) + \text{AVG}([\text{Quality of Materials}]) + \text{AVG}([\text{Market Proficiency}]))$
Delivery Time	$\text{AVG}([\text{Delivery Time}]) / (\text{AVG}([\text{Environment}]) + \text{AVG}([\text{Delivery Time}]) + \text{AVG}([\text{Service Level}]) + \text{AVG}([\text{Price}]) + \text{AVG}([\text{Quality of Materials}]) + \text{AVG}([\text{Market Proficiency}]))$
Service Level	$\text{AVG}([\text{Quality of Materials}]) / (\text{AVG}([\text{Environment}]) + \text{AVG}([\text{Delivery Time}]) + \text{AVG}([\text{Service Level}]) + \text{AVG}([\text{Price}]) + \text{AVG}([\text{Quality of Materials}]) + \text{AVG}([\text{Market Proficiency}]))$
Price	$\text{AVG}([\text{Price}]) / (\text{AVG}([\text{Environment}]) + \text{AVG}([\text{Delivery Time}]) + \text{AVG}([\text{Service Level}]) + \text{AVG}([\text{Price}]) + \text{AVG}([\text{Quality of Materials}]) + \text{AVG}([\text{Market Proficiency}]))$
Quality if Material	$\text{AVG}([\text{Quality of Materials}]) / (\text{AVG}([\text{Environment}]) + \text{AVG}([\text{Delivery Time}]) + \text{AVG}([\text{Service Level}]) + \text{AVG}([\text{Price}]) + \text{AVG}([\text{Quality of Materials}]) + \text{AVG}([\text{Market Proficiency}]))$
Marketing Proficiency	$\text{AVG}([\text{Market Proficiency}]) / (\text{AVG}([\text{Environment}]) + \text{AVG}([\text{Delivery Time}]) + \text{AVG}([\text{Service Level}]) + \text{AVG}([\text{Price}]) + \text{AVG}([\text{Quality of Materials}]) + \text{AVG}([\text{Market Proficiency}]))$

Willingness to pay for each feature:

A willingness to pay was calculated for each attribute of each feature to see how the change in each attribute contributed to the change in price of the product.

To keep the calculation a bit simpler, we created a field Utility Change by \$1, which basically told us how much the price of the product would be impacted in \$ with negative values showing it will reduce and positive values showing it will increase. Since the change in price was being calculated, the Price feature was removed from this calculation.

Environmental Friendliness	WTP – 30%	$\text{AVG}([30\% \text{ CO2 reduction}]) * [\text{Utility Change by } \$1]$
	WTP – 50%	$\text{AVG}([50\% \text{ CO2 reduction}]) * [\text{Utility Change by } \$1]$
Delivery Time	WTP – 21	$\text{AVG}([21 \text{ Days}]) * [\text{Utility Change by } \$1]$
	WTP - 30	$\text{AVG}([30 \text{ Days}]) * [\text{Utility Change by } \$1]$
Service Level	WTP – Warranty and Maintenance	$\text{AVG}([5\text{-Year Warranty \& Free Maintenance}]) * [\text{Utility Change by } \$1]$
	WTP – Warranty, Maintenance and Upgrade	$[5\text{-Year Warranty, Free Maintenance \& Installation, \& Upgradeability}]$
Quality if Material	WTP – Above Average	$\text{AVG}([A \text{ bit higher than market average}]) * [\text{Utility Change by } \$1]$
Marketing Proficiency	WTP -Proficient	$\text{AVG}([Very \text{ proficient and have good communication skills}]) * [\text{Utility Change by } \$1]$

Principal Component Analysis

The initial data preparation stage involved loading essential packages and datasets. Subsequently, non-numerical columns were purged from the data. To ensure equal weighting across attributes during the Principal Component Analysis (PCA), a scaling operation was applied using the `Scale=TRUE` parameter within the R code. The PCA then decomposed the cleaned data, and the resulting components, along with their corresponding singular values, loadings, and the proportion of variance explained (PVE) for each component, were captured. These extracted metrics facilitated the identification of the most influential attributes shaping customers' perceptions.

```
#computations specifically asked within question
# Compute singular values
singular_values <- pca$sdev
singular_values
pca_singular_values <- cbind(attribute, `Singular Values` = singular_values)

# Compute loadings
loadings <- pca$rotation
pca_loadings <- cbind(attribute, `Loadings` = loadings)

# Compute Proportion of Variance Explained
pve <- (singular_values^2) / sum(singular_values^2)
pca_pve <- cbind(attribute, `Proportion of Variance Explained` = pve)
```

Furthermore, a dimensionality reduction technique known as Principal Component Analysis (PCA) was leveraged to generate a perceptual map. This map visually depicted the relative positioning of product models within the reduced dimensional space defined by the principal components. The extracted data, encompassing both component scores and model labels, were subsequently exported to comma-separated values (CSV) files for further analysis and presentation purposes.

Perceptual Mapping

This code performs a Principal Component Analysis (PCA) on perceptual data about different models (brands or products). It analyses the attributes used to create a perceptual map visualizing how consumers perceive these models.

First, the code identifies the relevant attributes and calculates factor scores for the two most important principal components. These scores represent the influence of each attribute on these components. Then, it computes additional informative metrics: singular values (importance of each component), loadings (correlation between attributes and components), and the proportion of variance explained by each component. All this information is stored in separate data frames for further analysis.

Finally, the code focuses on the models themselves. It calculates normalized scores for each model on the two principal components. These scores essentially represent the position of

each model on the horizontal and vertical axes of the perceptual map. By plotting these scores, you can visualize how different models are perceived relative to each other based on the chosen attributes. This perceptual map provides valuable insights into consumer preferences and helps identify potential competitive advantages.

Results and Discussion:

Conjoint Analysis

The part-worths for each respondent were estimated using R, utilizing the *caPartUtilities()* function. Within this function, the intercept and coefficients were adjusted based on the baseline case. Figure 2 displays the part-worth results.

```
> print(part.worths)
```

	intercept	0% CO2 reduction	30% CO2 reduction	50% CO2 reduction	14 days	21 days	30 days	5-year warranty
OP1	1.798	0	-1.500	-2.000	0	-1.000	-0.500	0
OP2	1.723	0	-1.666	-2.000	0	0.834	0.000	0
OP3	0.611	0	-2.167	-3.167	0	1.000	0.667	0
OP4	0.078	0	-0.667	-0.167	0	-2.000	-1.333	0
OP5	0.510	0	-1.667	-2.167	0	1.000	-0.334	0
OP6	1.834	0	0.000	-1.000	0	-1.666	-2.333	0
OP7	1.034	0	-1.167	-1.833	0	0.667	-0.166	0
OP8	-0.435	0	-2.167	-2.333	0	-1.667	-2.833	0
OP9	-0.012	0	-2.166	-3.000	0	-2.000	-2.667	0
OP10	-0.067	0	0.334	-2.833	0	-0.500	-1.500	0

	5-year warranty & free maintenance	5-year warranty, free maintenance and installation, & upgradeability	1000 GBP
OP1	0.333	-0.834	0
OP2	1.500	-1.667	0
OP3	-0.500	-2.833	0
OP4	0.167	0.000	0
OP5	-1.167	-2.167	0
OP6	-0.333	-2.166	0
OP7	0.167	-1.166	0
OP8	-1.667	-0.334	0
OP9	0.666	0.666	0
OP10	0.500	-1.000	0

	1200 GBP	1500 GBP	Market average	A bit higher than market average	Not very proficient and poor communication
OP1	0.500	-0.500	0	-1.076	0
OP2	1.000	-0.667	0	-4.124	0
OP3	0.500	1.167	0	-3.626	0
OP4	-0.334	-2.000	0	-2.824	0
OP5	-0.500	1.166	0	-2.726	0
OP6	1.000	0.000	0	-2.000	0
OP7	-0.833	-0.166	0	-4.300	0
OP8	-0.333	0.333	0	-3.226	0
OP9	1.000	0.333	0	-2.650	0
OP10	-0.834	-2.667	0	-2.276	0

	Very proficient and have good communication skills
OP1	-1.326
OP2	-0.874
OP3	-1.376
OP4	-1.576
OP5	-1.476
OP6	-1.000
OP7	-0.300
OP8	-0.976
OP9	-2.150
OP10	-0.524

Figure 3: Part Worth Values

Think of the intercept as the starting point (their utility) when none of the features have any impact (the base level). The numbers in these columns show how much each feature, at its different levels, affects this starting point (intercept). A negative number means that feature level makes the person like it less (disutility) compared to the intercept. A positive number means that feature level makes them like it more (utility) compared to the intercept.

In other words, these numbers tell you how much each feature, at its different strengths, adds to (positive) or takes away from (negative) someone's initial liking (represented by the

intercept) based on the absence of any feature influence. Market segmentation aided by conjoint analysis allows a product strategy to be created that meets consumer wants and increases market share for retailers. (Green, P.E. and Krieger, A.M., 1991, Wind, Y.J. and Bell, D.R., 2008)

Visualizations

Average Attribute Importance

The average attribute importance plot in figure 4 represents how much each attribute was important for the customers when they considered the product. As per the image, it seems the most important feature was quality of material, and the market proficiency contributed the least.

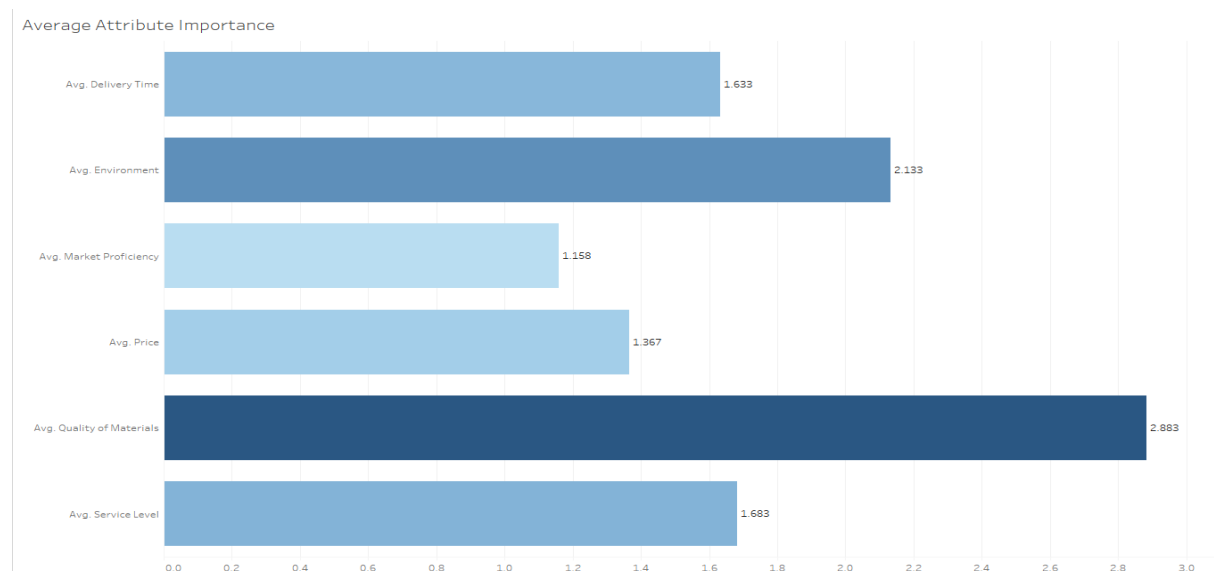


Figure 4: Average Attribute Importance

% Average Attribute Importance

Figure 5 shows the above calculation in the form of percentages.

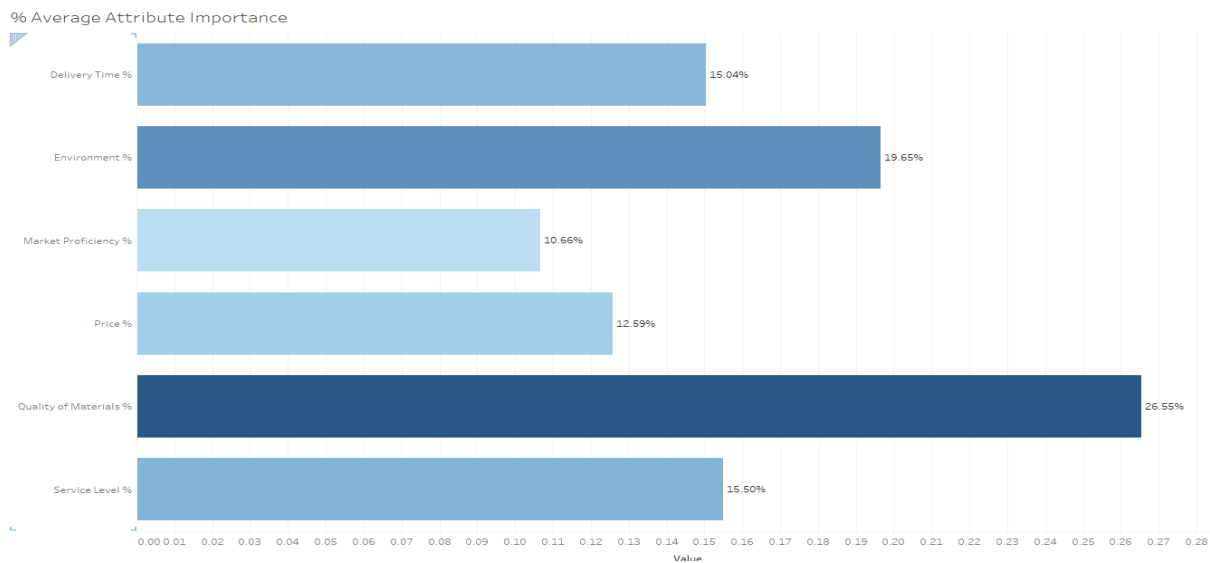


Figure 5: % Average Attribute Importance

Average WTP for a feature

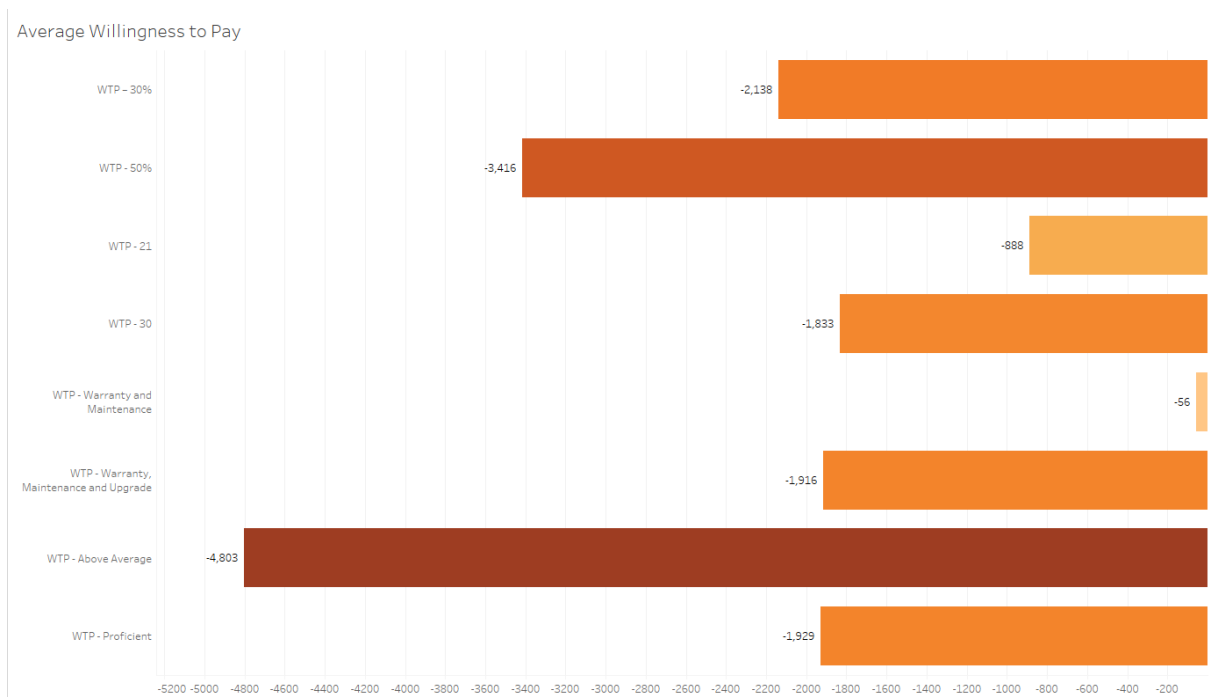


Figure 6: Willingness to Pay for a feature.

From the Willingness to Pay plotted above, we could see the following results for each feature and its attribute.

- Environmental Friendliness:** Customers value a 0% CO2 reduction and would pay \$2,138.12 less to accept a product with only a 30% reduction. Similarly, they can be expected to pay \$3,415.53 lesser for a product with 50% CO2 reduction compared to the base model.

- **Delivery Time:** Customers are willing to pay an extra \$888.37 to receive their product within the baseline delivery time (14 days) compared to waiting 21 days. Moreover, the difference more than doubles as the customers want to pay \$1,832.56 less for a 30-day delivery time. This suggests a significant preference for faster delivery.
- **Service Level:** The customers are more likely to only pay the minimum price for the warranty and would pay \$55.65 less to get a warranty along with free maintenance. Similarly, the customer would further pay \$1916.19 less compared to base 5-year warranty if upgradation option is also added.
- **Marketing Proficiency:** Interestingly, customers would prefer a discount of \$1,929.02 for a product with less marketing proficiency. This might indicate a preference for less marketing influence.
- **Quality:** Customers seem to highly value quality. A significant discount of \$4,803.07 is expected to move from "higher than average" quality to just "average".

Principal Component Analysis

Principal Component Analysis (PCA) is relied upon in present conjoint analysis to identify the most important attributes influencing consumer purchases. Through PCA, client priorities for features can be identified. The relative significance of each attribute and the amount of aggregate data variance explained by each major component are determined by singular values, loading factors, and PVEs (Mishra, et al., 2017).

Singular Values:

The singular values, representing the amount of variation explained by each principal component, are stored in a vector named *singular_values* in decreasing order. The singular values are observed to decrease sequentially. This indicates that each subsequent principal component captures progressively less variance. Consequently, the first principal component explains a greater proportion of variance compared to the others.

```
> singular_values
[1] 2.5706809 1.6280258 0.7919579 0.5192277 0.4727061 0.4599958 0.3677798 0.3505730 0.2775728 0.2281128 0.1484736
```

Factor Loading:

The order of the factor loadings reveals how attributes relate to each principal component.

If we look at the results, the value which is highest in the column represents the most impact on the principal component. For example, feature 3 impacts PC1 the most, Similarly, Feature 1 impacts PC7 the most since it is the highest value in the column.

Moreover, it needs to be kept in mind that the positive values represent positive loading and negative value means negative loading.

> loadings

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11
Feature1	-0.3625305	-0.01612440	0.22574419	-0.022540255	-0.10284468	0.10879743	-0.367723810	0.754091423	0.235701617	-0.13928524	0.124895628
Feature2	0.3739160	-0.04374371	0.17531118	-0.002591838	-0.05848381	-0.16855369	-0.057277736	0.230824925	0.054035270	0.84641949	0.140695441
Feature3	0.3681852	0.04932413	0.06148414	0.256607885	-0.39399530	0.33616451	-0.214303077	-0.001142134	0.198427848	-0.04937979	-0.660606481
Feature4	0.3300569	-0.24878402	-0.14001476	-0.067676157	-0.54004744	-0.07143563	0.001495989	0.222358441	-0.575830072	-0.24782351	0.256492062
Feature5	-0.2941514	-0.27469408	-0.16118879	0.854828743	-0.07732727	-0.24449705	-0.021119857	-0.032193501	-0.046901228	0.10149369	0.039530246
Feature6	0.3461033	0.14303825	-0.34181851	0.245899314	0.07502912	0.46493964	0.020668302	0.008571929	0.359498251	-0.09439426	0.567448697
Feature7	-0.2004563	0.46337482	-0.40316904	0.068076532	0.16466591	0.33048032	-0.050010522	0.231840021	-0.528377185	0.27067295	-0.181361780
Feature8	-0.3065113	0.23164699	-0.42881517	-0.214848616	-0.59953955	-0.19401702	0.265780836	-0.025935128	0.358582624	0.15903909	-0.008414634
Feature9	-0.2349429	-0.42941765	0.20576657	-0.030462908	-0.08978128	0.57081745	0.587305101	0.059746952	-0.047403982	0.17778541	-0.029823537
Feature10	-0.2069162	-0.46234863	-0.28977993	-0.264690521	-0.04832960	0.24356284	-0.605097617	-0.336150240	-0.001735039	0.21382515	0.053507085
Feature11	0.2140177	-0.41357106	-0.52854459	-0.126789179	0.36131875	-0.18352168	0.174603192	0.395629107	0.170640677	-0.07225950	-0.319594676

Proportion of Variance (PVE):

The proportion of variance explained (PVE) for each principal component is calculated by dividing each unique singular value by the sum of all singular values. The image depicts the Proportion of Variance Explained (PVE) values for each principal component. PVE represents the fraction of the total variance in the data explained by each component.

> pve

```
[1] 0.600763659 0.240951627 0.057017934 0.024508858 0.020313737 0.019236011 0.012296544 0.011172858 0.007004241 0.004730495 0.002004037
```

Perceptual Mapping

Figure 7 presents a perceptual map created using Tableau. This map visualizes how different product models are perceived based on two key factors. It reveals that Model 30 is positioned close to Feature 11, suggesting a strong association between this feature and the perception of that particular model. Conversely, Models 3 and 5, are located away from all features on map, are less influenced by extreme values of any specific features.

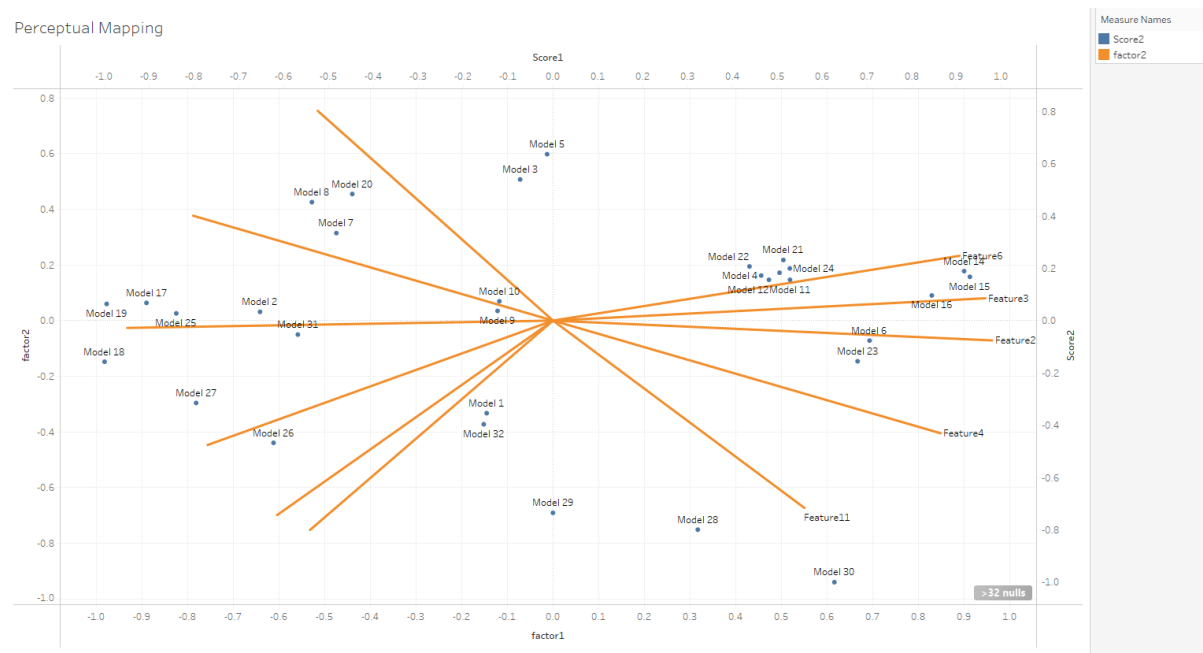


Figure 7: Perceptual Map

Conclusion:

This study highlights the effectiveness of combining conjoint analysis, PCA, and perceptual mapping to understand consumer preferences for product attributes. By analyzing the influence of individual features on choices, the study identified distinct consumer segments with shared priorities.

These insights are valuable for businesses to optimize products and marketing strategies. Tailoring offerings and messaging to each segment can enhance customer satisfaction and loyalty. Precise segmentation also allows for efficient resource allocation and effective go-to-market strategies. Ultimately, aligning product innovation with consumer desires leads to a competitive advantage.

This approach benefits both product development and marketing efforts. By targeting messages and channels that resonate with specific segments, businesses can maximize their marketing ROI. Overall, the study provides a framework for understanding consumer preferences, leading to actionable insights for strategic decision-making, ultimately driving sales, customer satisfaction, and market success.

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R Code:

CONJOINT ANALYSIS

```
library(conjoint)

library(dplyr)

# Read the Excel file and store it in a Laptop_DP
product_profiles <- read.csv(file.choose())
product_profiles <- product_profiles[,-1]
pref_data <- read_excel("C:/Users/Uzair Khan/Downloads/Conjoint Preferences.xlsx")
#set.seed(40385187) Seed is not required at this step since the dataset is already available.
pref_data <- pref_data[,-1]

## Set up attributes and levels as a list
attrib.level <- list(
  Environmental.friendliness = c("0% CO2 reduction", "30% CO2 reduction",
    "50% CO2 reduction"),
  Delivery.time = c("14 days", "21 days", "30 days"),
  Service.Level = c("5-year warranty", "5-year warranty & free maintenance",
    "5-year warranty, free maintenance and installation, & pgradeability"),
  Price = c("1000 GBP", "1200 GBP", "1500 GBP"),
  Quality.of.material = c("Market average", "A bit higher than market average"),
  Marketing.Proficiency = c("Not very proficient and poor communication",
    "Very proficient and have good communication skills"))

## Check for correlation in fractional factorial product profiles
print(cor(caEncodedDesign(product_profiles)))
summary(product_profiles)

## Run the conjoint analysis study

## Set up attributes and levels as a vector and Estimate the part-worths for each respondent
attrib.vector <- data.frame(unlist(attrib.level,use.names=FALSE))
colnames(attrib.vector) <- c("levels")
part.worths <- NULL
```

```

for (i in 1:ncol(pref_data)){
  temp <- caPartUtilities(pref_data[,i], product_profiles, attrib.vector)

  ## Pick the baseline case

  Adjust coding as needed based on number of attributes and levels

  Base Case: Environmental.friendliness 0% CO2 reduction, Delivery.time 14 days,
  Service.Level 5-year warranty, Price 1000 GBP,
  Quality.of.material Market average, Marketing.Proficiency Not very proficient and poor communication"

  Base_Environmental.friendliness <- temp["0% CO2 reduction"]; Base_Delivery.time <-
temp["14 days"]; Base_Service.Level <- temp["5-year warranty"]; Base_Price <-
temp["1000 GBP"]; Base_Quality.of.material <- temp["Market average"];
Base_Marketing.Proficiency <- temp["Not very proficient and poor communication"]

  ## Adjust Intercept

  temp["intercept"] <- temp["intercept"] - Base_Environmental.friendliness -
Base_Delivery.time - Base_Service.Level - Base_Price - Base_Quality.of.material -
Base_Marketing.Proficiency

  ## Adjust Coefficients

  ## Environmental.friendliness

  L1 <- length(attrib.level$Environmental.friendliness) + 1 ## Add 1 for the intercept
  for (j in 2:L1){temp[,j] <- temp[,j] - Base_Environmental.friendliness}

  ## Delivery.time

  L2 <- length(attrib.level$Delivery.time) + L1
  for (k in (L1+1):L2){temp[,k] <- temp[,k] - Base_Delivery.time}

  ## Service.Level

  L3 <- length(attrib.level$Service.Level) + L2
  for (l in (L2+1):L3){temp[,l] <- temp[,l] - Base_Service.Level}

  ## Price

  L4 <- length(attrib.level$Price) + L3
  for (m in (L3+1):L4){temp[,m] <- temp[,m] - Base_Price}

  ## Quality.of.material

  L5 <- length(attrib.level$Quality.of.material) + L4
  for (n in (L4+1):L5){temp[,n] <- temp[,n] - Base_Quality.of.material}

```

```

## Marketing.Proficiency

L6 <- length(attrib.level$Marketing.Proficiency) + L5

for (o in (L5+1):L6){temp[,o] <- temp[,o] - Base_Marketing.Proficiency}

part.worths <- rbind(part.worths, temp)
}

rownames(part.worths) <- colnames(pref_data)

print(part.worths)

## Export part-worths from analysis

write.csv(part.worths, file.choose(new=TRUE), row.names = FALSE) ## Name the file
conjoint_partworth

*** PRINCIPAL COMPONENT ANALYSIS***

#Installing and loading pacakge

install.packages("data.table")

library(data.table)

set.seed(40385187)

#Reading the perception data

perception.data <- read.csv(file.choose())

summary(perception.data)

#Removing first column since all columns needs to be numeric

new_data <- perception.data[,2:length(perception.data)]

#Getting the Principal Component Scores on Perceptions

pca_score <- prcomp(new_data, retx = T, scale = T)

## Perceptual Map Data - Attribute Factors and CSV File

attribute <- as.data.table(colnames(new_data)); setnames(attribute, 1, "Attribute")

factor1 <- pca_score$rotation[,1]*pca_score$sdev[1]; factor2 <-
pca_score$rotation[,2]*pca_score$sdev[2]; path <- rep(1,nrow(attribute))

pca_score_factors <- subset(cbind(attribute, factor1, factor2, path), select = c(Attribute,
factor1, factor2, path))

pca_score_origin <- cbind(attribute, factor1 = rep(0,nrow(attribute)), factor2 =
rep(0,nrow(attribute)), path = rep(0,nrow(attribute)))

```

```

pca_score_attributes <- rbind(pca_score_factors, pca_score_origin)

write.csv(pca_score_attributes, file = file.choose(new=TRUE), row.names = FALSE) ## Name
file perceptions_attributes.csv

#computations specifically asked within question

# Compute singular values

singular_values <- pca_score$sdev
singular_values

pca_score_singular_values <- cbind(attribute, `Singular Values` = singular_values)

# Compute loadings

loadings <- pca_score$rotation
loadings

pca_score_loadings <- cbind(attribute, `Loadings` = loadings)

# Compute Proportion of Variance Explained

pve <- (singular_values^2) / sum(singular_values^2)
pve

pca_score_pve <- cbind(attribute, `Proportion of Variance Explained` = pve)

## Perceptual Map Data - Model Factors and CSV File

score1 <- (pca_score$x[,1]/apply(abs(pca_score$x),2,max)[1])
score1

score2 <- (pca_score$x[,2]/apply(abs(pca_score$x),2,max)[2])
score2

pca_score_scores <- subset(cbind(perception.data, score1, score2), select = c(Model, score1,
score2))

write.csv(pca_score_scores, file = file.choose(new=TRUE), row.names = FALSE) ## Name file
perceptions_scores.csv

```